

# Shahmun Jafri

shahmun.z.jafri@gmail.com | [shahmun.com](https://shahmun.com) | [linkedin.com/in/shahmunjafri](https://linkedin.com/in/shahmunjafri)

## EDUCATION

### University of California, San Diego

San Diego, CA

*B.S. in Mathematics-Computer Science*

*Sep 2022 – Jun 2026*

- Relevant Coursework: System Programming, Data Science and Optimization, Machine Learning, Advanced Data Structures and Algorithms, Operating Systems, Computing Theory

## EXPERIENCE

### Software Engineering Intern

Jun 2025 – Sep 2025

*Inprintz*

*Sunnyvale, CA*

- Architected an order system with **Next.js 15**, **TypeScript**, and **Prisma ORM**, saving **80%** data entry time.
- Designed a normalized **PostgreSQL** schema with **Prisma 6** modeling **5+** entity relationships.
- Built **10+** RESTful API routes via **Next.js** Server Actions with a **35MB** upload limit for print-ready assets.
- Containerized with **Docker**, cutting environment setup time by **70%** across dev and production.
- Eliminated processing errors via end-to-end type-safe validation with **TypeScript** and **Prisma**.
- Leveraged **Next.js** SSR and dynamic routing for real-time order status updates.

### Undergraduate Researcher

Jul 2023 – Jul 2024

*Zaidi Lab, SJSU*

*San Jose, CA*

- Engineered an **Arduino**-based plasma device in **C++**, cutting resource use by **15%** and firmware size by **30%**.
- Collaborated with a research team to design and validate embedded control systems across **50+** test cycles.
- Increased energy efficiency by **25%** by designing a custom air-cooled aluminum radiator.
- Documented experimental results and contributed to lab reports supporting ongoing research publication.
- Collected and logged sensor data across **30+** trials to validate device performance and inform design iterations.

## PROJECTS

### LLM Impact on Stack Overflow | *Python, SQL, Pandas, NLP, Statistics*

- Queried **6 years** of Stack Overflow data (**1,007** observations, **14** tags) via **SQL** to study LLM-driven shifts.
- Applied **DiD**, **Mann-Whitney U-tests**, and **survival analysis** — **67%** vs. **46%** collapse post-ChatGPT.
- Built a composite **NLP** complexity scorer revealing easy questions vanished from LLM-friendly boards.
- Led a team of **5** to design the study, divide analysis tasks, and present findings to faculty.

### Patient Journey Intelligence | *Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Jupyter*

- Engineered an **ETL pipeline** classifying **3.5M+** patient encounters into **4** types averaging **194-day** cycles.
- Trained a **Gradient Boosting** classifier with **Scikit-learn** on EHR and SDoH features.
- Surfaced **51%** complexity rates in metabolic syndrome and **10x** longer cycles in chronic vs. resolved cases.
- Collaborated with a team of **4** across a 48-hour hackathon to deliver a full analysis and presentation.

### Southern California Wildfire Project | *Python, NumPy, Seaborn, Pandas, GeoPandas, Folium*

- Led a team of **4**, processing **~100K** records (**CSV + GeoJSON**) and engineering features for a clean dataset.
- Created **geospatial heatmaps** with **Folium** to identify high-risk wildfire regions by seasonal factors.
- Documented a reproducible pipeline (cleaning → modeling → reporting) with ethics and bias considerations.
- Applied regression modeling to identify structural and environmental predictors of wildfire damage risk.

### MNIST Digit Classifier | *Python, NumPy, Scikit-learn, Matplotlib*

- Built a binary digit classifier on **70,000** MNIST images using gradient descent over **10,000** iterations.
- Implemented least-squares linear regression from scratch in **NumPy** to classify digits 0 vs. 1 with **0.2%** test error.
- Tuned step size and stopping criteria, achieving convergence with **0.1%** training error and **0.2%** test error.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, JavaScript/TypeScript, Java, HTML/CSS, Intel x86 Assembly, Golang

**Frameworks:** React, Next.js, Node.js, Prisma

**Developer Tools:** Git, Docker, PostgreSQL, GitHub, Valgrind, Vim, VS Code

**Libraries:** NumPy, Pandas, Seaborn, Matplotlib

**Certifications:** Stanford Machine Learning Certification

## VOLUNTEER WORK

- Taught CS concepts to elementary students using **Scratch** to promote early STEM engagement.